

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 900+315

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – VUP

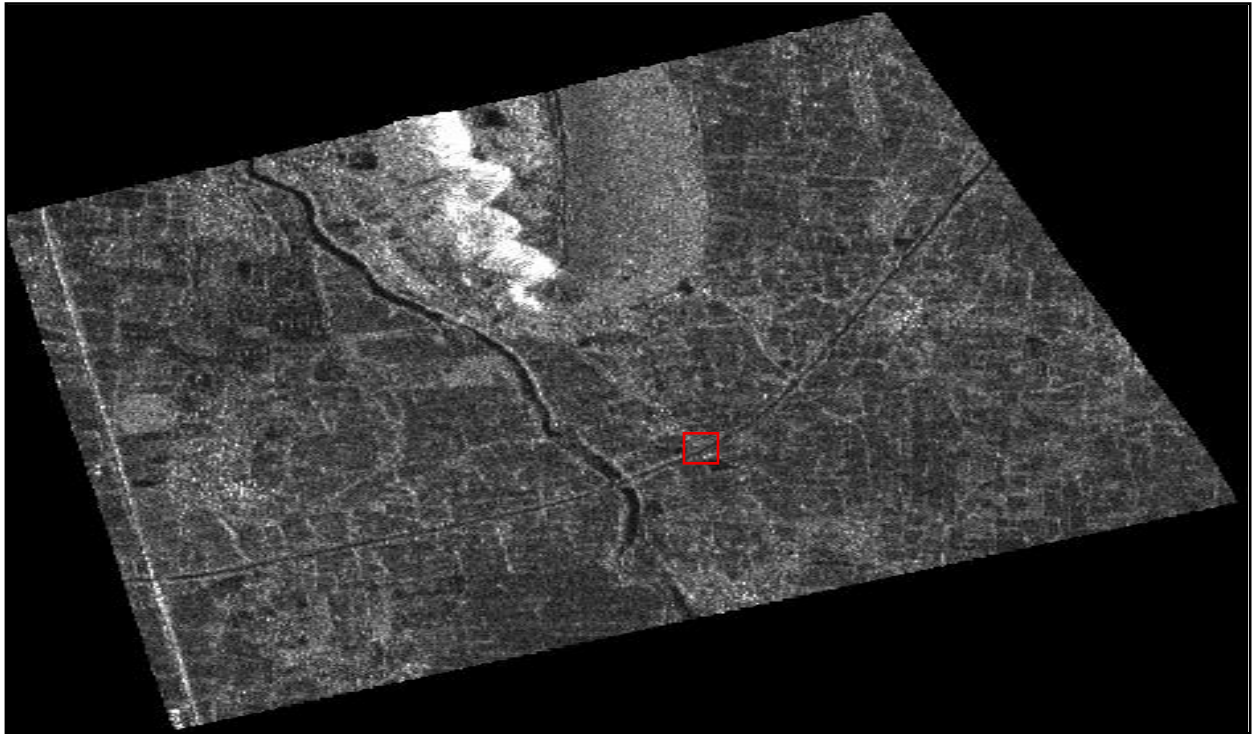
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

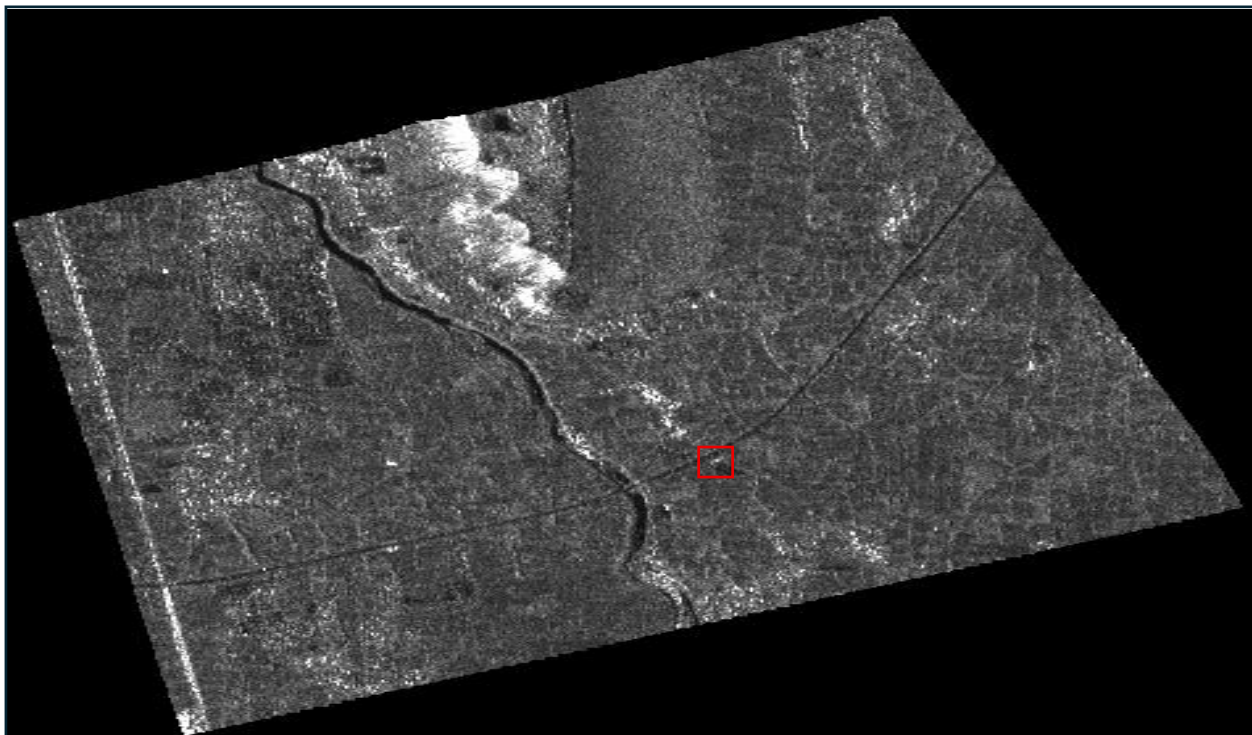
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

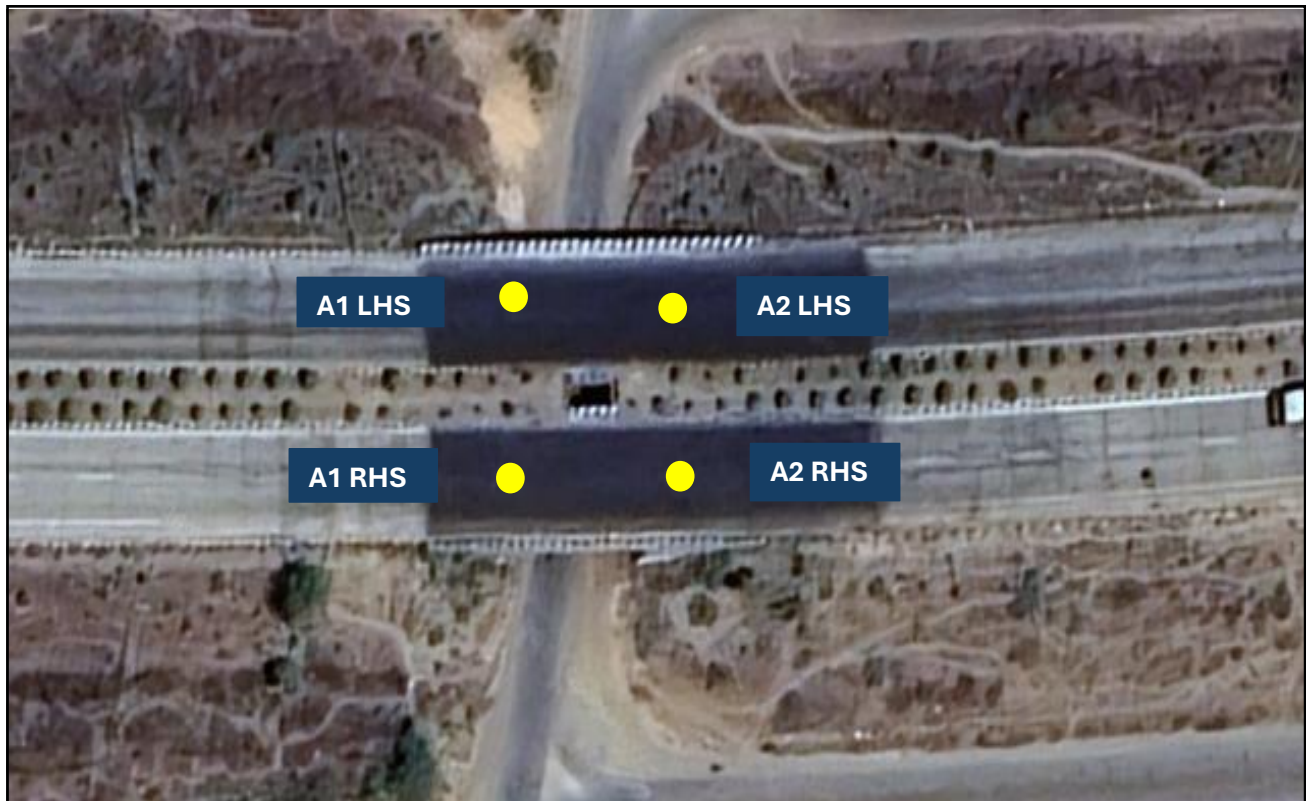


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

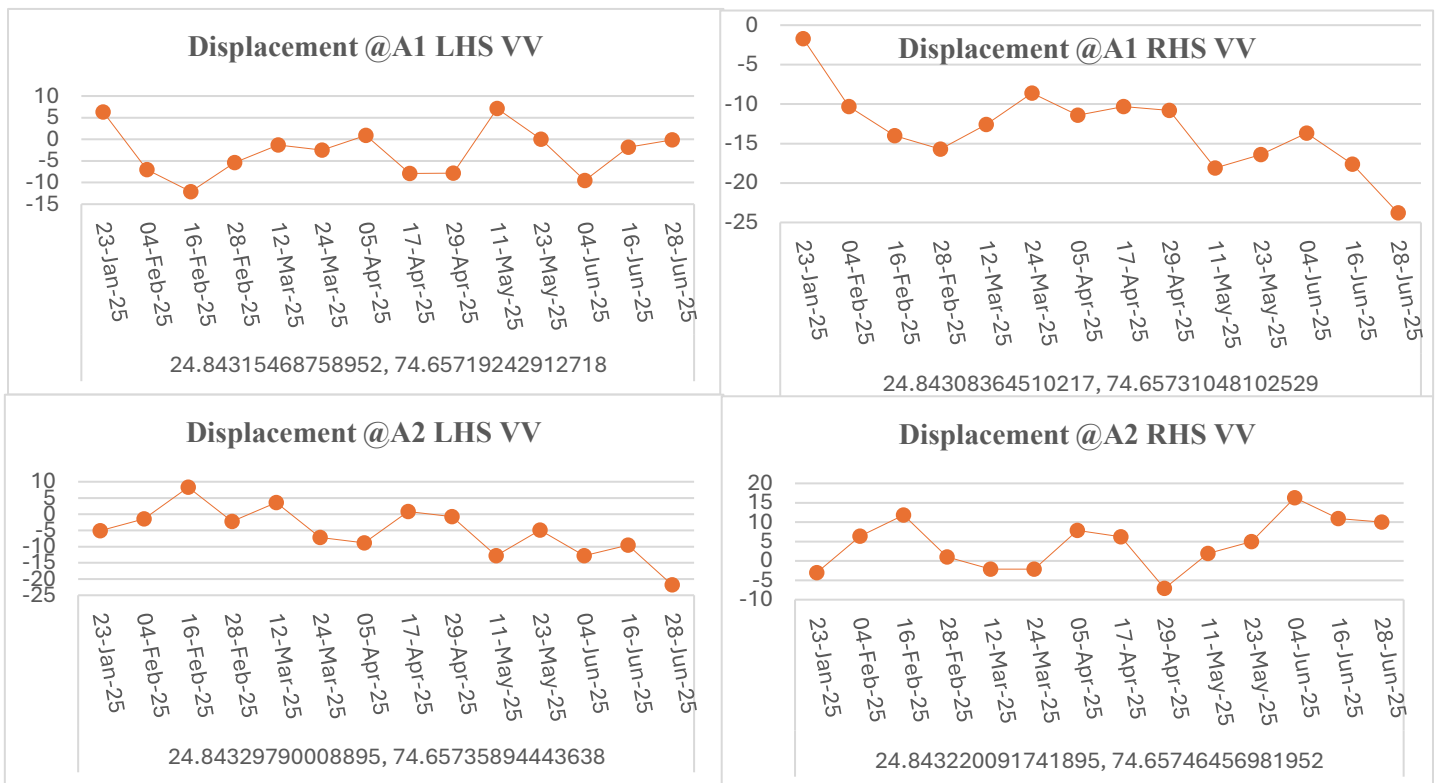
VUP – CH-900+315

Length – 9.10M

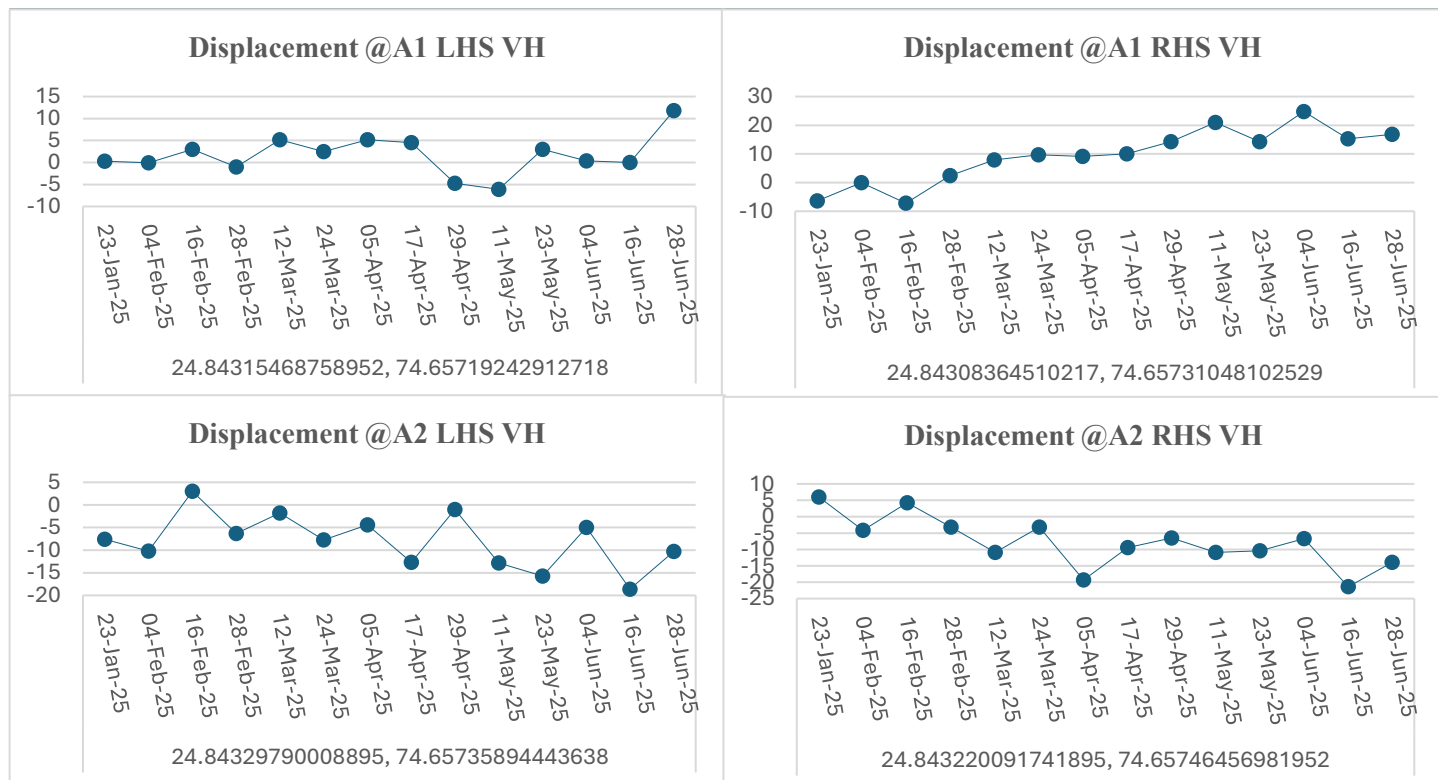


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



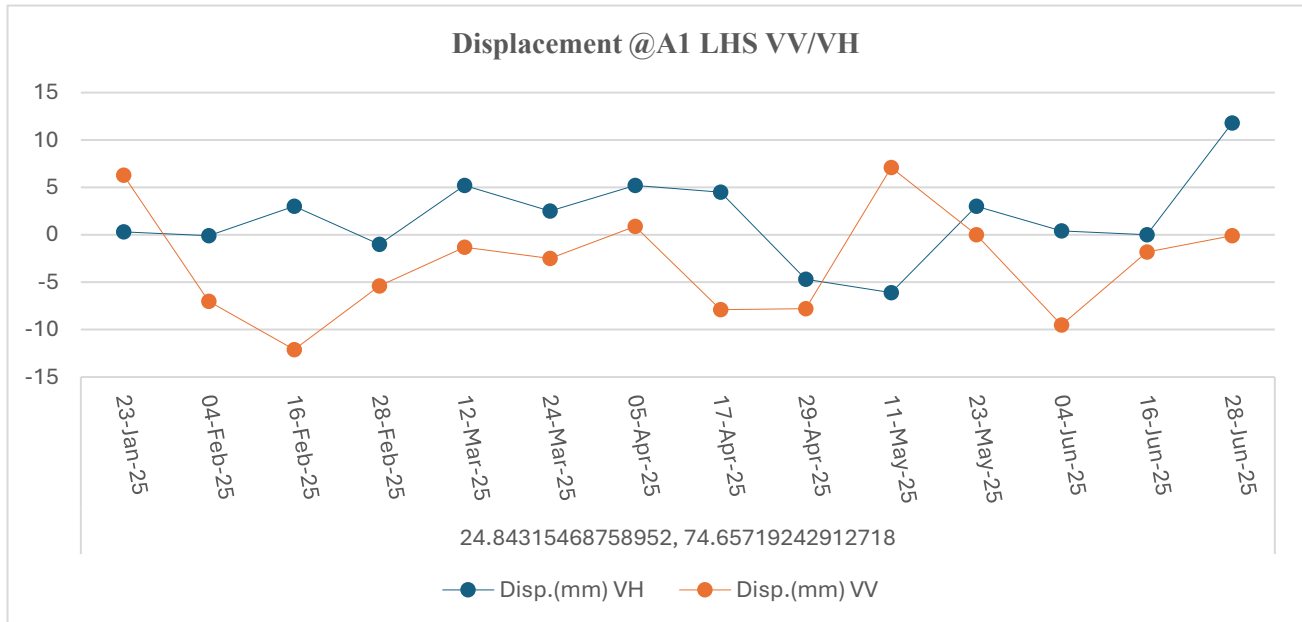
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.84315468758952, 74.65719242912718)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** ranges approximately from -12 mm to +7 mm, showing moderate variability with repeated cycles of settlement and uplift.
- **Horizontal Displacement (VH)** at A1 LHS fluctuates around the neutral axis, ranging approximately from -6 mm to +12 mm.

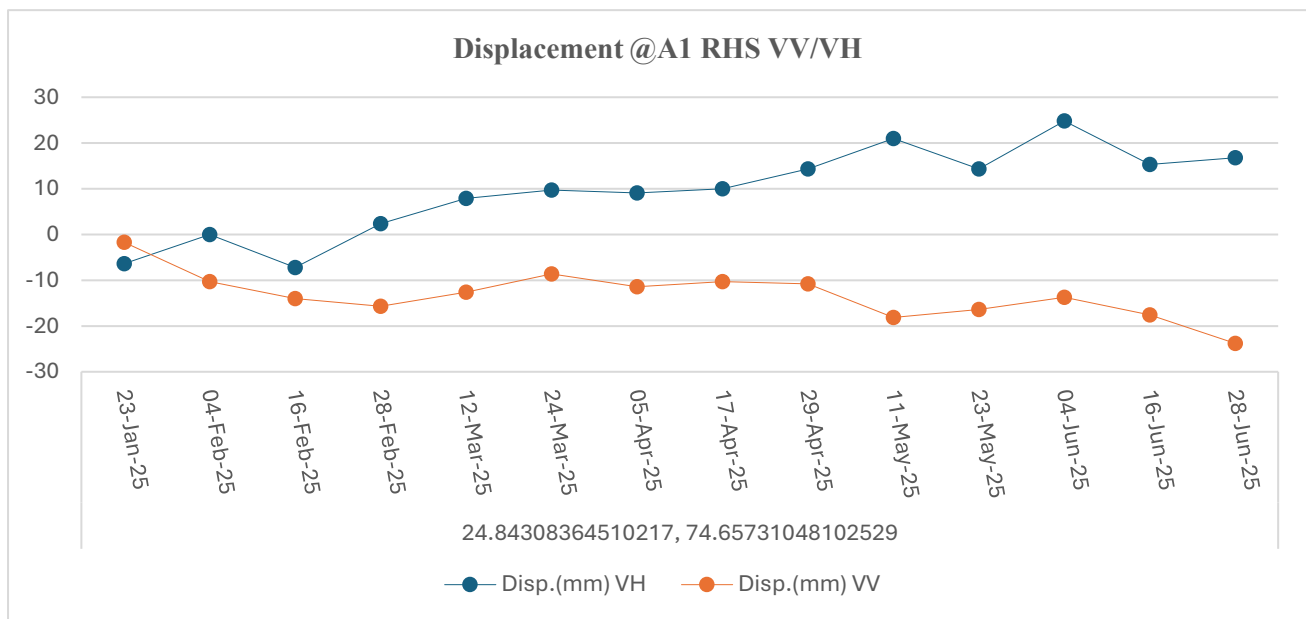
Critical Events & Anomalies:

- **February 2025:** VV records a pronounced downward movement (-12 mm) while VH remains mildly positive, suggesting localized vertical settlement without significant lateral response.
- **April 2025:** Both VH and VV drop into the negative range, with VH around -5 mm and VV near -8 mm, marking a short-term combined settlement phase.
- **June 2025:** VH recovers to a maximum positive value while VV stabilizes close to 0 mm, suggesting lateral recovery with vertical stabilization toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.84308364510217, 74.65731048102529)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A1 RHS shows a clear increasing positive trend, ranging approximately from –6 mm to +25 mm.
- **Vertical displacement (VV)** Vertical displacement remains consistently negative, ranging approximately from –8 mm to –24 mm, reflecting sustained vertical settlement with limited recovery throughout the monitoring period.

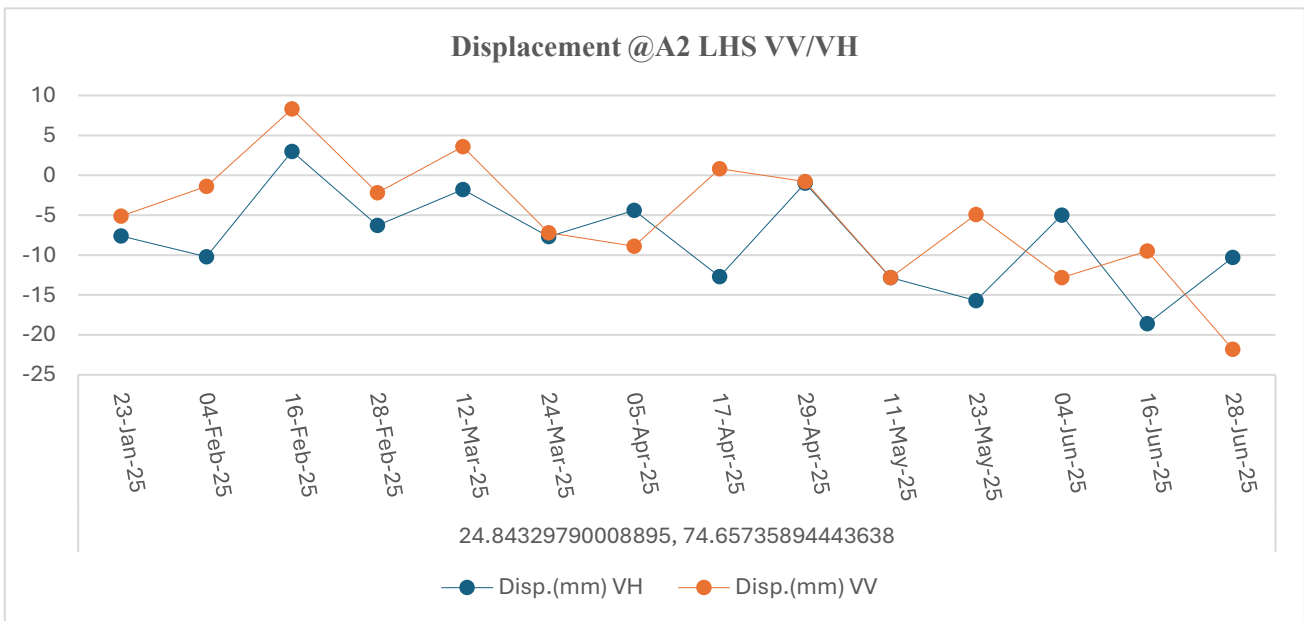
Critical Events & Anomalies:

- **February 2025:** VH transitions from slightly negative to positive, while VV deepens toward –15 mm, suggesting onset of lateral movement concurrent with vertical settlement.
- **March 2025:** VH stabilizes around +8 to +10 mm, whereas VV fluctuates between –10 mm and –13 mm, indicating steady lateral displacement under ongoing settlement
- **June 2025:** VH remains elevated and stable, while VV records its maximum settlement, suggesting long-term vertical settlement with lateral displacement having largely stabilized.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.84329790008895, 74.65735894443638)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A2 LHS ranges approximately from +3 mm to –18 mm, showing a predominantly negative trend over time.
- **Vertical Displacement (VV)** reflecting moderate to significant vertical settlement.

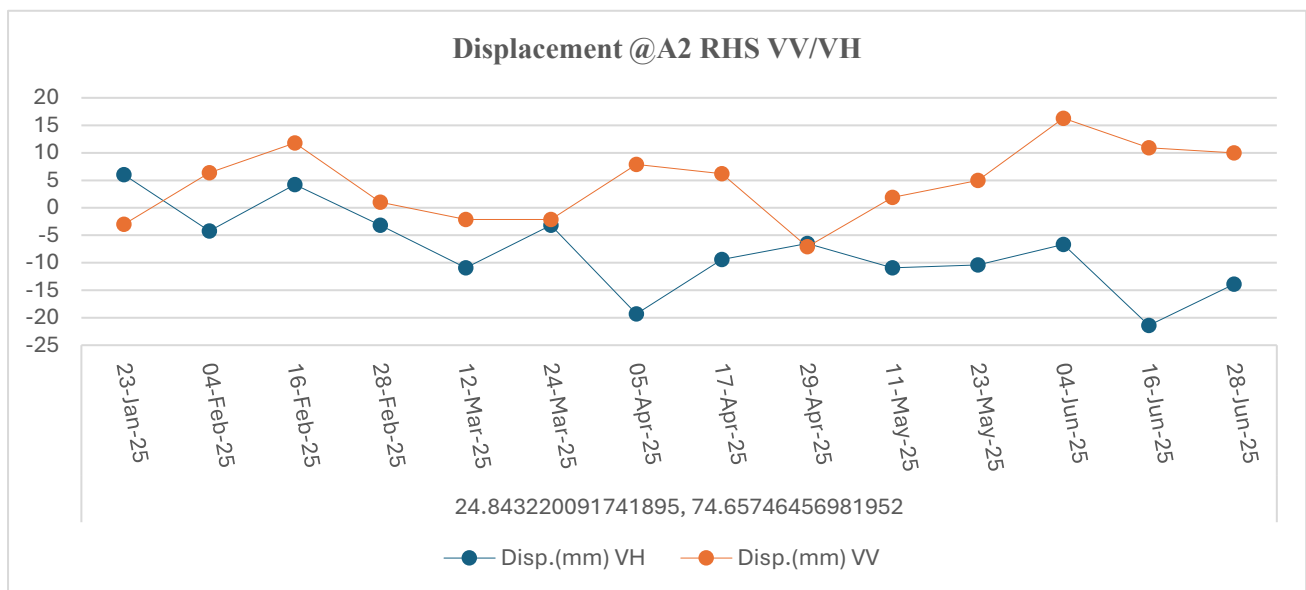
Critical Events & Anomalies:

- **February 2025:** VV records a positive peak alongside a short-lived positive VH, suggesting temporary uplift and lateral recovery, possibly due to transient loading or thermal effects.
- **March 2025:** VH and VV trend downward, with VH moving from about –2 mm to –12 mm and VV declining toward –8 mm, indicating combined lateral and vertical settlement onset.
- **June 2025:** Both VH and VV attain their maximum negative values indicating cumulative long-term lateral displacement and vertical settlement.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.843220091741895, 74.65746456981952)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement varies from approximately -4 mm to +16 mm, showing moderate vertical variability.
- **Horizontal (VH) displacement** at A2 RHS ranges approximately from +6 mm to -22 mm. The response is predominantly negative, with intermittent short-term recoveries, indicating overall lateral movement biased toward the negative direction over the monitoring period.

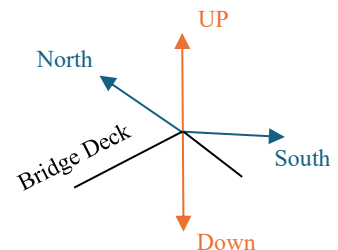
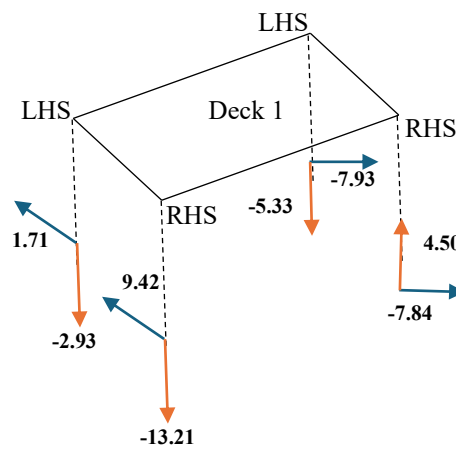
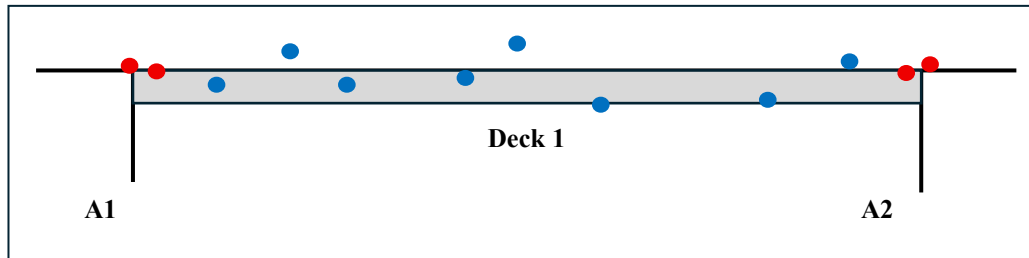
Critical Events & Anomalies:

- **February 2025:** VH briefly shifts into the positive range (+5 mm) while VV rises to about +10 mm.
- **March 2025:** VH declines sharply to around -10 mm, while VV drops slightly negative, indicating the onset of combined lateral and vertical settlement behaviour.
- **June 2025:** VH records another major negative peak as VV attains its maximum positive value, indicating significant lateral movement occurring simultaneously with vertical uplift, a critical condition warranting close observation.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits predominantly downward vertical displacement, indicating overall settlement-dominated behaviour across the deck. The maximum downward displacement is observed at the lower central deck region, reaching approximately -13.2 mm, representing the most significant vertical response in this configuration.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear directional variability, reflecting asymmetric and torsional behaviour of the deck slab. The central lower deck region experiences the largest horizontal displacement, reaching approximately +9.4 mm, indicating dominant lateral movement concentrated near the mid-deck support

CONCLUSION:

- The bridge deck slab demonstrates a non-symmetric deformation pattern, dominated by vertical settlement combined with torsional horizontal response. Overall, the observed displacement behaviour reflects differential support response and asymmetric stiffness distribution, warranting continued monitoring and correlation with loading and boundary conditions.